



ATM Software

XFS4IoT SP-Dev Workgroup

2nd June 2026

- Recap previous meeting
- Embedded systems experiments
- TLS cross-platform support
- Next meeting



Recap meeting March 2026

- Framework update
 - Framework version 3.3.0 is available
 - SP template package version 3.3.0 is available
- Recap of the German DK protocol
- Guest speaker - Synaforce
 - PoC and demo

Embedded systems experiments

Recap of previous experiments

- Ran an adapted version of the SP-Dev framework on a Meadow F7 board:
- Meadow.OS, runs full .NET standard v2.1
 - STM32F7 MCU
 - 32MB RAM (2MB available)
 - 64MB Flash storage



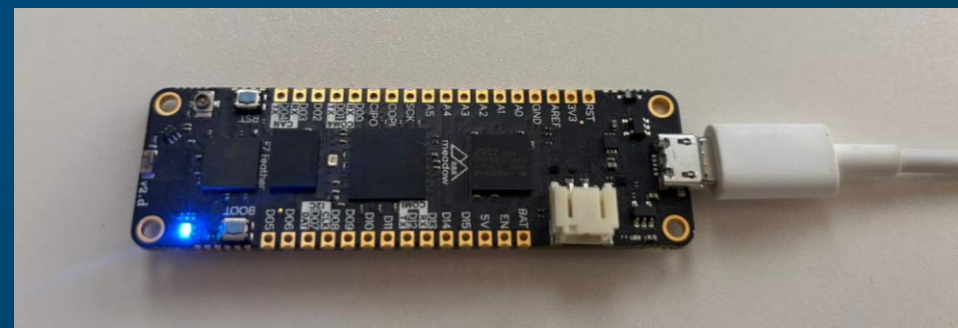
Main Framework changes	Main results
• Framework migrated to .NET standard v2.1 • Use of Websocket-sharp-netstandard-2.1 • Compiled with AOT, native code • Meadow tool failed to 'trim' the project	• Serialization/deserialization caches significant memory • Only one command could be executed (e.g. SetLight) • A second command fails to allocate enough memory

Main Framework changes

- Tried JsonTypeInfo => OOM. Kept Type + JsonSerializerContext
- Removed double JSON decoding for header sniffing & GenericMessageContext: reduced allocations in half
- Above also allowed to trim the code
- Moved the Service Publisher out from the board
- Minimized Status & Capabilities structures

Main results

- Could execute multiple instances of the 3 main commands for a Lights SP:
 - Capabilities, Status, SetLight (could control the on-board LED)





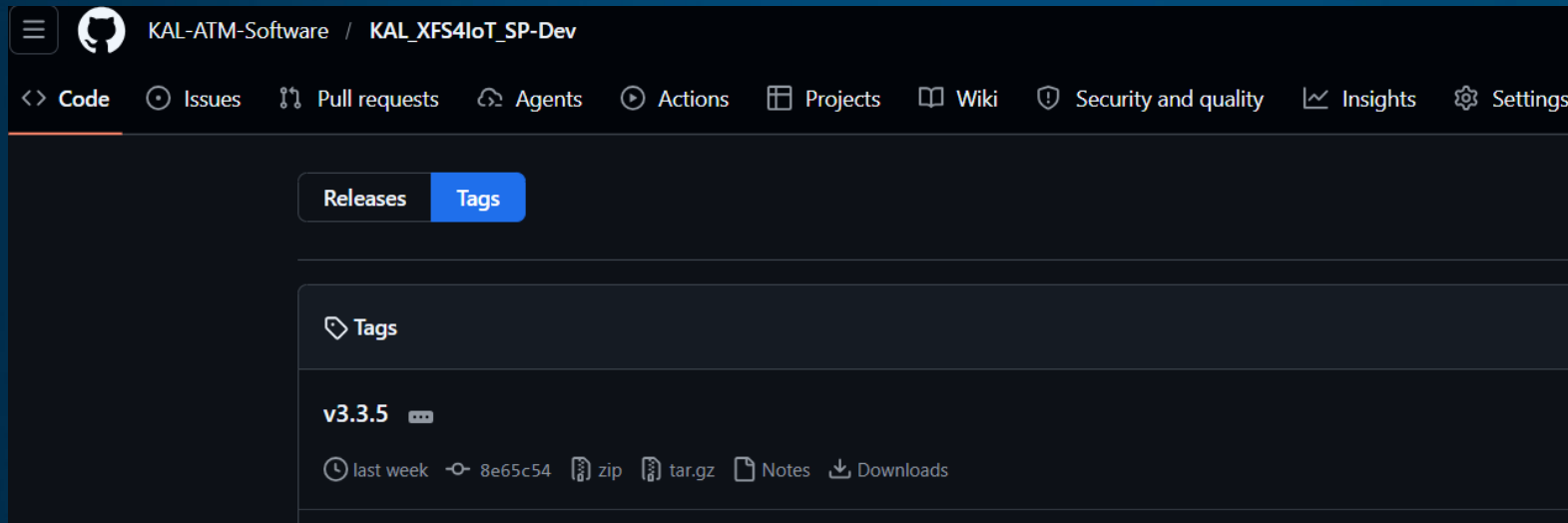
-  Latest version supports Generics
-  Latest version supports libraries for WebSockets and JSON
-  Latest version does NOT support Tasks or Async/Await code
-  "Virtual Device" does not support network (no WebSockets)



TLS cross-platform support

- Supported TLS on Windows only
 - The framework uses an `HttpListener` object to handle incoming connections
 - `HttpListener` relies on `http.sys`, a Windows-only kernel component, so it is not supported on other platforms such as Linux, Android or iOS
- Supported TLS on cross-platform
 - Replaced `HttpListener` with `TcpListener` + `SslStream`
 - The `TcpListener` is wrapped to upgrade an incoming HTTP/HTTPS connection to a WS/WSS WebSocket protocol
 - `SslStream` handles the entire TLS handshake with the client, including key exchange, cipher negotiation and establishing the secure session

- Framework update for supporting TLS cross-platform
 - Framework version 3.3.5 is available
 - SP template package version 3.3.5 is available



Steps	Results
• Created a self-signed certificate in Ubuntu • Installed it on a Windows machine • Compiled CardReader SP sample for Linux and ran it in Ubuntu • Specified the certificate path and password on the SP config • Used TestClientForms from Windows to connect with the SP	• Successfully established a TLS communication • Commands were processed correctly • Confirmed encryption with WireShark



TestClientForms

CardReader

CashDispenser

Text Terminal

Encryptor

PinPad

Printer

Lights

Auxiliaries

VendorMode

VendorApplication

BarcodeReader

Biometric

CashAccepter

Camera

CheckS

Service URI

wss://192.168.1.140

Port Number

443

CardReader URI

wss://192.168.1.140/xfs4iot/v1.0/SimCardReader/

Service Discovery

Device status

Online

Media Status

NotPresent

Device type

AcceptCard

EjectCard

CaptureCard

Capabilities

EMVClassQueryApplications

EMVClassPerformTransaction

Status

[20/05/2026 19:40:38.513] ServicePublisher.GetServices - command - 2

[20/05/2026 19:40:38.628] ServicePublisher.GetServices - completion - 2

[20/05/2026 19:40:38.645] Common.Capabilities - command - 3

[20/05/2026 19:40:39.049] Common.Capabilities - completion - 3

[20/05/2026 19:42:08.461] Common.Status - command - 4

[20/05/2026 19:42:09.593] Common.Status - completion - 4

```
{
  "header": {
    "name": "Common.Status",
    "requestId": 4,
    "type": "completion",
    "version": "3.0",
    "completionCode": "success"
  },
  "payload": {
    "common": {
      "device": "online",
      "devicePosition": "inPosition",
      "powerSaveRecoveryTime": 0
    },
    "cardReader": {
      "media": "notPresent",
      "chipPower": "noCard",
      "chipModule": "ok",
      "magWriteModule": "ok"
    }
  }
}
```

No.	Time	Source	Destination	Protocol	Length	Info
...	1.346699140	192.168.1.140	192.168.1.136	TLSv1.3	628	Application Data, Application Data
...	1.347103829	192.168.1.136	192.168.1.140	TLSv1.3	255	Application Data
...	1.347520912	192.168.1.140	192.168.1.136	TLSv1.3	205	Application Data
...	1.349518774	192.168.1.136	192.168.1.140	TLSv1.3	186	Application Data
...	1.351201575	192.168.1.140	192.168.1.136	TLSv1.3	170	Application Data
...	1.364833626	192.168.1.136	192.168.1.140	TCP	54	57735 → 443 [ACK] Seq=1129 Ack=3286 Win=6
...	1.398940532	192.168.1.136	192.168.1.140	TCP	54	57736 → 443 [ACK] Seq=1135 Ack=3015 Win=6
...	1.560950295	192.168.1.140	192.168.1.136	TLSv1.3	2...	Application Data
...	1.561421501	192.168.1.136	192.168.1.140	TCP	54	57736 → 443 [ACK] Seq=1135 Ack=5917 Win=6
...	1.600277588	192.168.1.136	192.168.1.140	TCP	66	[TCP Port numbers reused] 57730 → 5853 [S
...	1.600278177	192.168.1.136	192.168.1.140	TCP	66	[TCP Port numbers reused] 57725 → 5847 [S
...	1.600278203	192.168.1.136	192.168.1.140	TCP	66	[TCP Port numbers reused] 57726 → 5849 [S
...	1.600310756	192.168.1.136	192.168.1.140	TCP	66	[TCP Port numbers reused] 57728 → 5851 [S
...	1.600323232	192.168.1.140	192.168.1.136	TCP	54	5853 → 57730 [RST, ACK] Seq=1 Ack=1 Win=0
...	1.600325977	192.168.1.140	192.168.1.136	TCP	54	5851 → 57728 [RST, ACK] Seq=1 Ack=1 Win=0

```

> Frame 75: Packet, 2956 bytes on wire (23648 bits), 00 15 5d 38 b3 0b 08 00 45 00 $...eM9...j8...E...
> Ethernet II, Src: Microsoft_38:b3:0b (00:15:5d:38:b3:0b) 40 06 f9 53 c0 a8 01 8c c0 a8 ~...@...@...S...
> Internet Protocol Version 4, Src: 192.168.1.140, Dst: 10.0.0.1 21 86 93 84 f6 c8 50 18 ... ..!...P...
> Transmission Control Protocol, Src Port: 443, Dst Port: 80 17 03 03 0b 51 b7 0e 27 3c 31 ...}... ..Q...<1...
> Transport Layer Security 67 3c c3 03 a1 18 bc b0 00 89 ... ..g<... ..
08 47 48 51 37 4b d2 ee c6 aa 8... ..) G HQ7K... ..
08 f1 f7 69 64 32 da 48 d0 9b ...r-5... ..id2-H...
38 03 1a 25 6c a7 27 2f 54 a5 ... ..k-8- %l-'/T...
e4 63 ce 41 72 18 cc a8 d1 8f ... ..!..c -Ar... ..
91 64 79 b2 4d 75 92 e1 cb bc r...I..d y-Mu... ..
35 66 56 be 5a 53 de ec 0d 86 ... ..5f V ZS... ..
b2 0b f5 14 da 74 45 fc 55 97 ...v...HC... ..tE-U...
65 f5 89 40 92 4e 63 03 72 d2 ...7e...e...@-Nc-r...
b7 2a a5 3d 17 83 a2 d1 c1 c1 KrJ... ..* ...=... ..
90 48 32 01 19 c0 27 e4 f4 3e W+$+...H 2... ..>...
dc 9e 58 46 03 81 fd b5 57 a5 ... ..<... ..XF... ..W...

```

```

":true,"off":false},"positions":{"exit":true,"transport":true},"cardTakenSensor":true}}
07:21:51.685 (037.197): Completed Common.Capabilities id:3
07:21:51.685 (037.197): Listening for new connections and on 2 existing connections
07:21:51.917 (037.429): Listening for new connections and on 1 existing connections
07:23:28.768 (134.280): Listening for new connections and on 2 existing connections
07:23:28.778 (134.290): Received:{"header":{"name":"Common.Status","requestId":4,"type":"command","version":"2.0","timeout":60000}}
07:23:28.778 (134.290): Sending: {"header":{"name":"Common.Status","requestId":4,"type":"acknowledgment","version":"2.0"}}
07:23:28.778 (134.290): Running Common.Status id:4
07:23:28.801 (134.313): Client Closed connection
07:23:28.801 (134.313): Listening for new connections and on 1 existing connections
07:23:29.384 (134.896): Sending: {"header":{"name":"Common.Status","requestId":4,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"common":{"device":"online","devicePosition":"inPosition","powerSaveRecoveryTime":0},"cardReader":{"media":"notPresent","chipPower":"noCard","chipModule":"ok","magWriteModule":"ok"}}}
07:23:29.384 (134.896): Completed Common.Status id:4

```

- CardReader compound devices can now be created without the storage interface, such as for a DIP card reader which does not support this class
- Addressed an issue reporting recognizable banknote items for the GetStorage command
- Updated conditions to send Storage.CountChangedEvent corresponding to 2023-3
- Added a way to notify updating cash unit counts from the device class to the framework. The device class can monitor count changes in the RunAsync method outside of XFS commands.

Zoom

Meeting once every quarter
at 1300 UK time for 30 mins

Next call: 1st September 2026
1300 UK, 0800 US EST, 2100 Tokyo time

Calls are 30 mins long

We will continue to use Zoom

(Interpretation in Japanese, Chinese and Spanish is available using Zoom's interpretation feature)